

```

6. def prepare_backtest_data(ticker, start_date, end_date, fields=None):
7.     """
8.     Prepare data for backtesting with proper adjustments and cleaning
9.
10.    Parameters:
11.    ticker (str): Stock symbol
12.    start_date (str): Start date in 'YYYY-MM-DD' format
13.    end_date (str): End date in 'YYYY-MM-DD' format
14.    fields (list): List of fields to keep, defaults to OHLCV
15.
16.    Returns:
17.    pd.DataFrame: Clean data ready for backtesting
18.    """
19.    # Set default fields if not provided
20.    if fields is None:
21.        fields = ['Open', 'High', 'Low', 'Close', 'Volume']
22.
23.    # Download data
24.    print(f"Downloading data for {ticker} from {start_date} to {end_date}...")
25.    data = yf.download(ticker, start=start_date, end=end_date)
26.
27.    # Verify we have data
28.    if len(data) == 0:
29.        raise ValueError(f"No data found for {ticker} in the specified date range.")
30.
31.    # Check for and handle missing values
32.    missing_values = data[fields].isna().sum().sum()
33.    if missing_values > 0:
34.        print(f"Found {missing_values} missing values. Handling missing data...")
35.        # Forward fill most recent values, then backward fill any remaining NAs
36.        data = data.fillna(method='ffill').fillna(method='bfill')
37.
38.    # Add adjusted fields if not present
39.    if 'Adj Close' in data.columns and 'Adj Close' not in fields:
40.        # Calculate adjustment ratio
41.        data['Adj Ratio'] = data['Adj Close'] / data['Close']
42.

```